

Connecting the dots

Solving NYT Connections

Automated Strategies for Semantic Grouping

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Project Overview

How can we solve Connections?

Siamese Network

Use two identical subnet weights to map words into valid clusters.

Static Embeddings

Use existing word vector maps and lexical databases to create an architecture that can map words into clusters.

The LLM

Use large language models like Gemini, Chat, and Claude to generate word groupings.

What is Connections

Defining Semantic Categories

Straightforward

Level: Easy

Common Groups

Level: Medium

Nuanced Association

Level: Hard

Abstract Connection

Level: Tricky

Given 16 words, group them into four groups of four of increasing difficulty

Siamese Network

Modeling Language with a limited scope

Model Overview

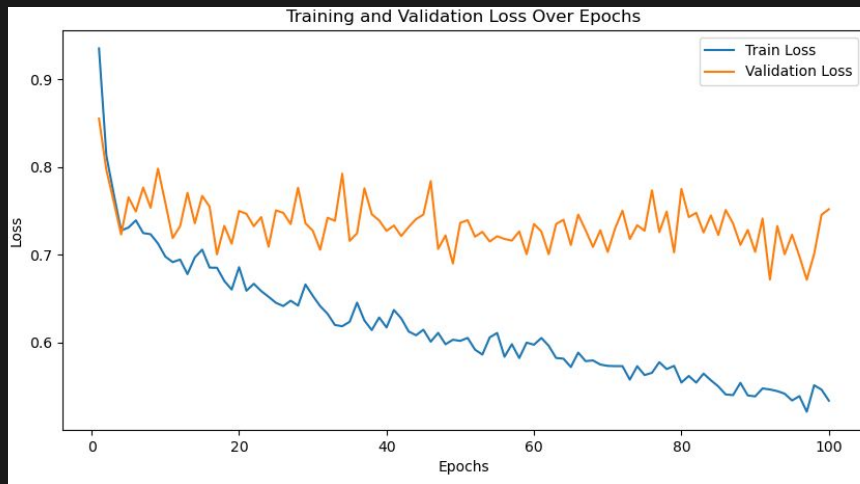
To go past pure semantic grouping, we need a secondary model to map the word play found in Connections. Using 1k past Connections game one model creates a vector space of a typical Connections game and becomes a space warper for an existing word transformer (MiniLM-L6-v2) that affects the weights created so clusterings that aren't connected by word definitions are captured.

Siamese Network Results

The ShortComings

Generalizing the nuances and possibilities of the natural language requires more than a few hundred clusterings.

Despite that it still got a 34% solve rate over 100 validation games



Siamese Network

Searching the network

Lookahead Beam search

Given already computationally expensive process of creating the 16×16 correlation matrix, for this method a heuristic beam search was implemented for a quicker search for optimal groupings.

The top 40 scoring groups found using average pairwise cosine distances and checks the next available highest scoring group for the top 40 and discards the lower scores until a grouping is found.

Siamese Network

Does a Siamese network actually help?

How much?

Running the same search with only the sentence transformer making the similarity matrix, the method only managed a solve rate of 1.34%.

```
Dataset Evaluated      : sorted_output.csv
Total Games Evaluated  : 915
Solve Rate             : 15 (1.64%)
Avg Groups Found/Game  : 0.46/ 4.00
```

Static Embeddings

Using existing Lexical Databases and similarity metrics

GloVe

Or Global Vectors for word representation, is a lexical database that contains the similarity of words through the probability of co-occurrence.

WordNet

Link word clusters by definitions and synonyms.

JellyFish

Supplement scoring with phonetic library.

Static Embeddings Methodology

Similarity matrices and search

Searching with greedy DFS

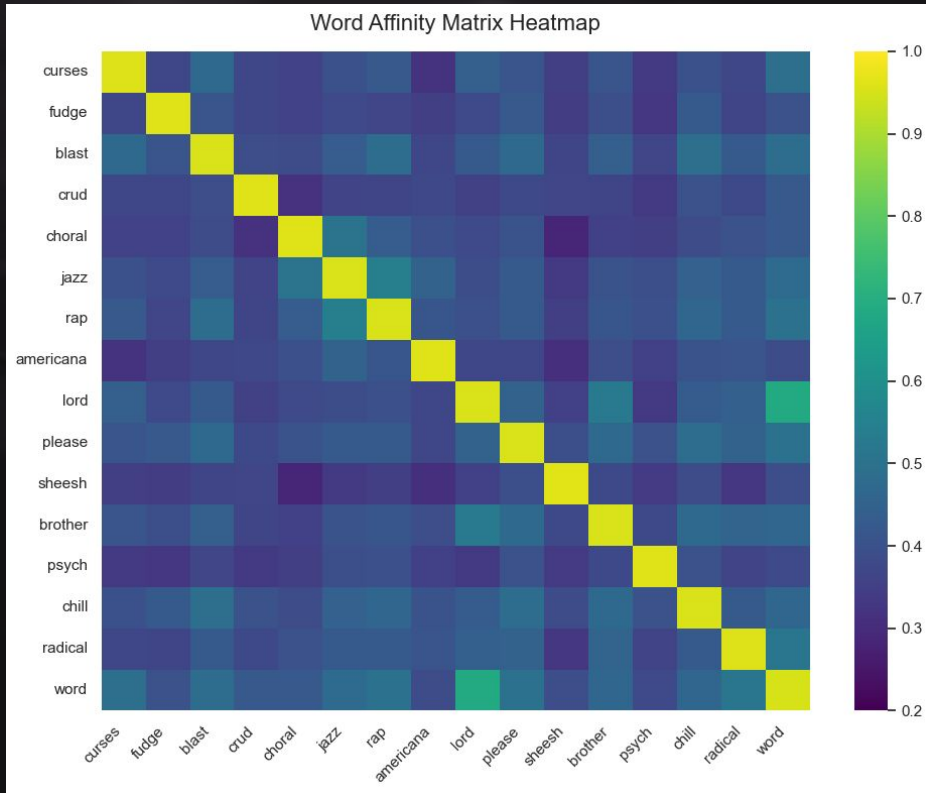
Each lexical database/library creates a 16x16 similarity matrix for a puzzle. How do we combine them?

Through iterating ranges of weights and alpha values on three thousand matrices...

optimal weights of 65% semantic (GloVe), 30% structural (WordNet), 5% orthographic (JellyFish)

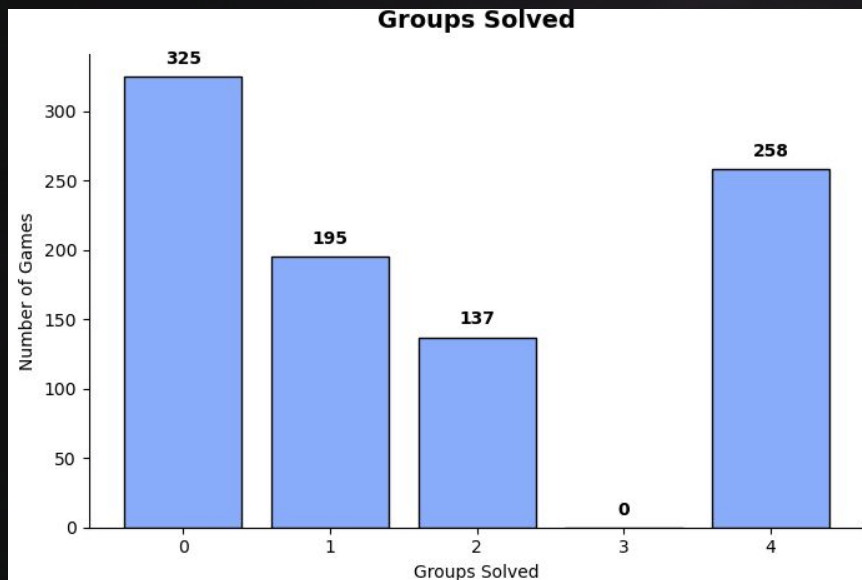
These are then combined with optimal weighting to form one matrix where 1,820 unique grouping of 4 words is calculated by score for greedy DFS.

Heatmap Matrix



Static Embeddings Results

The evolution of natural Language



Static Embeddings Results

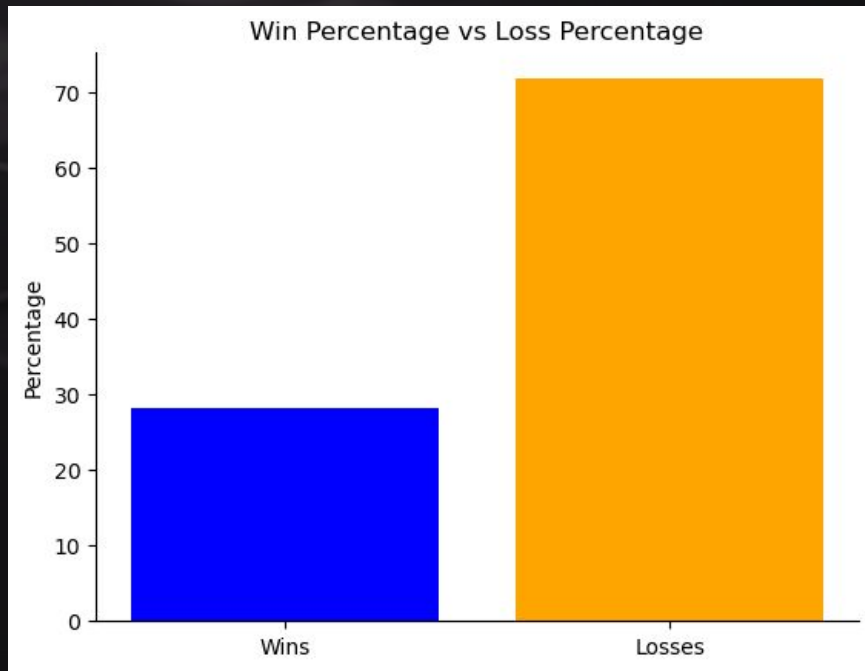
The evolution of natural Language

Win rate

Over the 915 past archived games, the system had a 28.1% solve rate.

Why

The static embeddings don't cover all possible connections a word can have. A category might be so obscure that the words have never been seen together in text, have no semantic similarity, and no structural similarity.



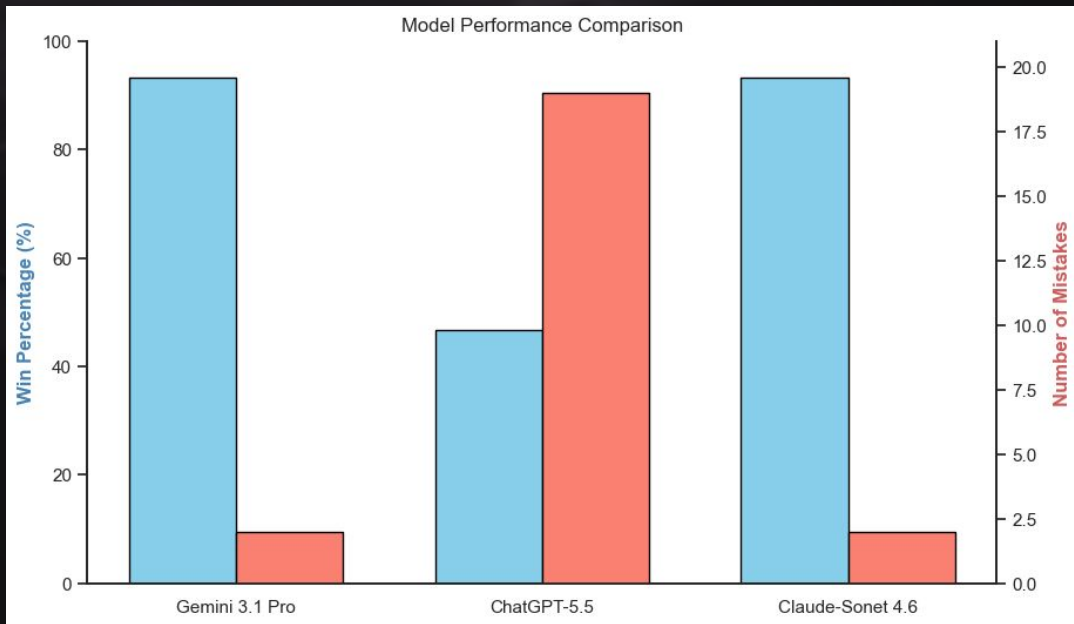
The LLM Approach

Using existing reasoning models

Results

Gemini and Claude performed similarly with few mistakes.

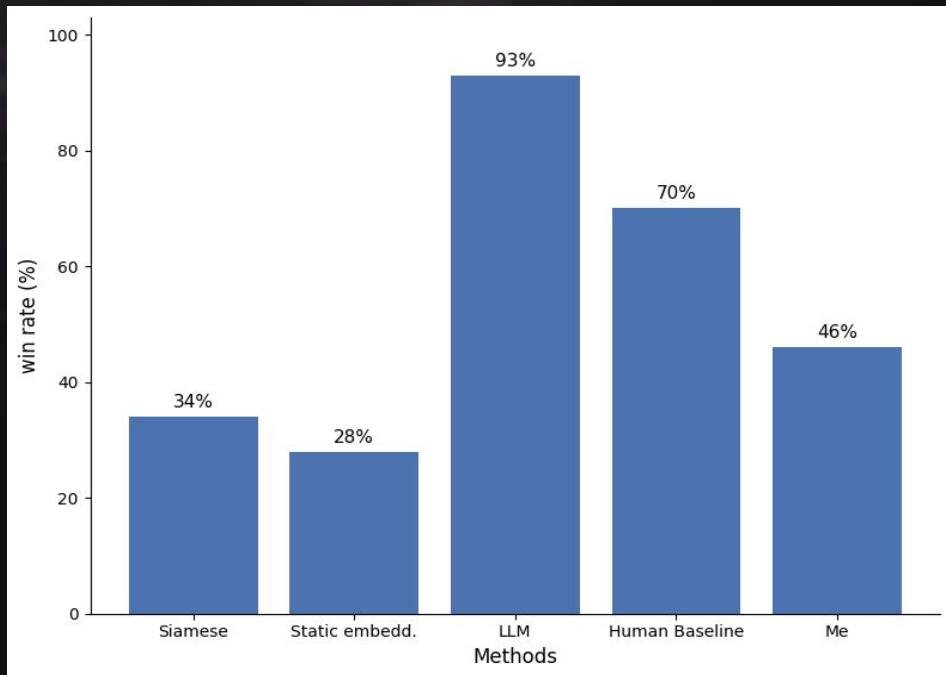
Chat-GPT on the other hand struggled on around half of the provided puzzles.



Comparison

How do we compare?

Unsurprisingly the LLM outperformed all other methods and the human baseline.



Today's Puzzle

FOREVER

SALON

COOK

SURGEON

NURSE

DOCTOR

FUDGE

SUPERFUDGE

CRY

BLUBBER

TROT

ALTER

FOUNDER

BABBLE

DEENIE

TEETHE

Today's Puzzle

Siamese Model

Lost

Static Embeddings

Lost

Gemini

Perfect Win

The Human

Lost

The BaseLine

58% with avg 2.4 mistakes

THINGS BABIES DO

BABBLE, CRY, NURSE, TEETHE

MODIFY DECEPTIVELY

ALTER, COOK, DOCTOR, FUDGE

JUDY BLUME BOOKS

BLUBBER, DEENIE, FOREVER, SUPERFUDGE

FISH MINUS A LETTER

FOUNDER, SALON, SURGEON, TROT

Improvements

Increasing accuracy and correctness

Siamese Model

The Siamese model only had 900 archived games to try to map the complexities of connections categories

A larger sentence transformer can also be used (mpnet-base-v2)

Static Embeddings

GloVe and Wordnet are good datasets but don't cover the relationships that can be found in Connections

The greedy DFS in most cases will find the highest scoring board but there exist cases where it returns a local minimum on one category. Global partitioning on nearly 2.6 million unique groupings will get the optimal group but with higher compute cost.

Conclusion

Difficulties of NLP

Connections is a hard game to solve without first solving the difficulties of Natural Language Processing.

Natural language is difficult to map onto pure vector distances since words can be polysemy with multiple different definitions, pronunciations, and context while having the same spelling.

This is furthered by addition of categories that go beyond vector semantics like anagrams and cultural context/knowledge.